Predicting Student Depression Using Machine Learning: *Comparing Ensemble Methods and Interpretable Linear Models*

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Declaration of Plagiarism

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Assignment Submission

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Prompts	• "Does this paragraph make sense and how can I improve it?" • "Can you help me reduce repetition in this section?" • "Can you condense this to the word count while keeping the meaning?" • "Improve the academic tone and flow of this paragraph." • "Where can this be clearer or more concise?" • "Is this machine learning evaluation approach correct?" • "Can you help me with this Python code error?" • "How should I interpret these model results?"

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List of Abbreviations

CGPA……..…..Cumulative Grade Point Average

CV.....Cross-Validation

GB…….Gradient Boosting

KNN…….K-Nearest Neighbour

LR……...……Logistic Regression

ML……..….....Machine Learning

RF…….....Random Forest

SHAP……...SHapley Additive exPlanations

SVM.....Support Vector Machines

VE……...Voting Ensemble

Abstract

Student depression is a growing public health concern requiring scalable approaches for early identification. This study applies supervised machine learning to a dataset of 27,901 university students to predict depression and identify key risk factors. Models evaluated include Logistic Regression (LR), Random Forest (RF), Gradient Boosting (GB), and a Voting Ensemble (VE), alongside SHAP-based interpretability and calibration analysis. All models achieved strong performance (ROC-AUC ≈ 0.92), with LR matching or slightly outperforming more complex methods. Key predictors include suicidal ideation, academic pressure, and financial stress. These findings support the use of interpretable models for transparent and effective early-warning systems.

1. Introduction

Depression among university students has become an increasingly important mental health challenge, driven by academic pressure, financial stress, and lifestyle-related factors (Aceijas et al., 2017). Early identification of at-risk individuals is critical for intervention; however traditional screening relies on self-reporting, limited by stigma and access constraints.

ML identifies patterns across psychological, academic, and demographic variables not captured by conventional approaches (Zulfiker et al., 2021; Vu et al., 2025). However, existing research is often limited by small or region-specific datasets and a strong reliance on classification metrics, with less attention given to model reliability, calibration, and generalisability.

To address these limitations, this study investigates the following research questions:

1. Which factors most strongly predict student depression?
2. Can ML models accurately classify depression using structured student data?
3. Do complex ensemble models outperform simpler linear models in this task?

A large-scale dataset of 27,901 students enables more robust evaluation. The full feature structure is provided in Appendix A. This work contributes to the literature by combining predictive performance, interpretability, and robustness evaluation within a unified framework for depression prediction.

2. Related Works

2.1 ML for Depression Prediction

ML is widely applied in mental health research to support the early identification of depression using behavioural, demographic, and psychosocial data. However, across studies, model performance is highly dependent on dataset quality, feature representation, and evaluation strategy. Consequently, reported results vary, particularly in robustness and generalisability. This suggests that performance differences in depression prediction are often influenced more by data characteristics than by modelling approaches alone.

2.2 Model Comparisons

Narayanrao and Lalitha Surya Kumari (2020) conducted a comparative analysis of decision trees, SVM, KNN, and Naive Bayes for depression classification. Their findings indicate that, although ML models can achieve strong predictive performance, no single algorithm consistently outperforms others across all conditions. Instead, model effectiveness is highly dependent on dataset characteristics.

This reflects a limitation in model comparison studies: performance differences are often driven by data structure rather than algorithm superiority. A further limitation is the reliance on accuracy-based evaluation, which is insufficient in healthcare contexts where different error types carry unequal consequences.

In such settings, metrics such as sensitivity, specificity, and probability calibration provide a more reliable understanding of model behaviour (Etemadi and Khashei, 2022), particularly given the risk of false negatives in clinical settings (Lavazza and Morasca, 2023). Earlier clinical research also supports this concern; for example, Klinkman et al. (1998) demonstrated that depression detection rates vary significantly depending on diagnostic definitions and measurement approaches, further underscoring the limitations of relying on single-metric performance metrics.

Collectively, these findings suggest that while traditional ML models can achieve strong classification performance, their reported effectiveness is strongly influenced by evaluation methodology, limiting the reliability of direct model comparisons across studies.

2.3 Feature-Based and Ensemble Methods

Zulfiker et al. (2021) applied six ML classifiers using socio-demographic and psychosocial features for depression prediction. Their best-performing model, AdaBoost combined with SelectKBest feature selection, achieved an accuracy of 92.56%.

However, the study is limited by a small sample size (n = 604), which increases the risk of overfitting and reduces generalisability. In addition, the dataset is region-specific, introducing potential cultural and demographic bias that limits external validity.

Although multiple classification metrics were reported (accuracy, precision, recall, F1-score, and ROC-AUC), the absence of calibration analysis and learning curve analysis restricts insight into model reliability and performance under varying data conditions (Mohr and van Rijn, 2024). This highlights a recurring gap in the literature, where strong classification performance is reported without sufficient assessment of model robustness.

2.4 Recent Advances

Vu et al. (2025) evaluated multiple ML models including LR, RF, Naive Bayes, SVM, and GB for depression prediction. Their results show strong predictive performance, with several models exceeding an AUC-ROC of 0.80.

A key finding is that LR performed competitively with more complex ensemble methods. This suggests that relationships between predictors and depression are largely linear and additive, which is consistent with evidence that structured healthcare datasets often exhibit dominant linear signal patterns rather than complex non-linear interactions (Lynam et al., 2020).

The study also employed SHAP to interpret feature contributions at both global and local levels, enhancing interpretability. However, evaluation remained focused on standard classification metrics such as accuracy, precision, recall, and AUC-ROC.

Across these studies, a consistent limitation emerges: while discrimination performance is well reported, calibration and learning curve analysis are rarely explored. As a result, there remains limited understanding of how reliably predicted probabilities reflect real-world risk, or how models perform under varying data conditions.

These findings indicate that increasing model complexity does not necessarily yield meaningful performance improvements, particularly in structured datasets where additive relationships dominate.

2.5 Research Gap

The existing literature demonstrates the strong potential for ML for depression prediction; however, several consistent gaps remain

First, many studies rely on small or region-specific datasets (e.g., Narayanrao and Lalitha Surya Kumari, 2020; Zulfiker et al., 2021), limiting generalisability and increasing the risk of sampling bias. This suggests that reported performance may not transfer reliably across populations.

Second, there is a strong emphasis on classification metrics such as accuracy, precision, recall, F1-score, and AUC-ROC. While these metrics capture discrimination performance, they do not fully reflect model reliability. Calibration analysis is necessary to assess probability trustworthiness (Silva Filho et al., 2023), while learning curve analysis helps evaluate model behaviour as training size increases and identify overfitting or underfitting (Deng et al., 2025).

Third, although recent work (Vu et al., 2025) incorporates interpretability techniques such as SHAP, there remains a lack of integrated evaluation frameworks that combine predictive performance with robustness and stability evaluation. Consequently, there is a limited understanding of how models perform under different data distributions and real-world deployment conditions.

Motivated by these limitations, this study adopts a large-scale dataset of over 27,000 student records to improve generalisability. In contrast to prior work, it integrates discrimination metrics, calibration analysis, learning curve diagnostics, and SHAP-based interpretability.

Furthermore, this study explicitly compares interpretable models such as LR with more complex ensemble methods to assess whether increased complexity provides meaningful performance gains in structured mental health prediction tasks.

3. Methodology

3.1 Dataset and Preparation

The dataset used in this study is the Student Depression Dataset compiled from multiple Indian universities. It contains 27,901 student records and 17 original variables, with a binary target variable indicating depression status (0 = non-depressed, 1 = depressed) (Appendix A). After removing three missing values in Financial Stress using deletion, the final dataset contained 27,898 records.

Preprocessing was conducted to ensure consistency across model inputs (see Appendix B for Python code). Binary variables were encoded as 0/1, ordinal variables were mapped to integer scales to preserve ordering, and nominal variables were transformed using one-hot encoding. This produced 105 encoded features. Numerical variables were standardised using z-score normalisation to ensure equal weighting. Although one-hot encoded variables were included in training, interpretation focused on the most influential predictors identified through SHAP and feature importance analysis, as high-cardinality dummy variables contributed minimally to overall predictive performance.

3.2 Classification Algorithms

Four supervised machine learning models were selected: LR, RF, GB, and a VE.

The selected models provide a structured comparison across varying levels of complexity and learning assumptions. LR serves as a linear baseline to test whether depression risk can be explained through additive feature relationships. RF introduces non-linear decision boundaries through bagging and feature randomness, allowing assessment of variance reduction in high-dimensional psychological data. GB is included as a sequential error-correcting method capable of capturing complex feature interactions. Finally, the VE integrates heterogeneous model outputs to evaluate whether aggregation improves stability beyond individual learners.

3.3 Feature Analysis and Interpretability

Rather than relying on a single method of feature evaluation, this study applies a multi-layered framework combining statistical and model-based approaches. These include correlation analysis, LR coefficients, feature importance, and SHAP values.

Correlation analysis was used to identify linear relationships between variables and depression status, providing an initial overview of feature relevance.

Model-based feature importance was conducted using SHAP analysis to quantify both global and local feature contributions using a theoretically grounded game-theoretic framework (Lundberg and Lee, 2017). This approach aligns with recent work in depression prediction that emphasises the importance of model interpretability for understanding feature influence in clinical contexts (Vu et al., 2025). Importantly, SHAP values are interpreted as contributions to model predictions rather than causal effects.

This approach reduces reliance on any single interpretability method and is instead validated across multiple perspectives.

3.4 Model Evaluation

Model performance was evaluated using accuracy, precision, recall, F1-score, and ROC-AUC to provide a balanced assessment of classification performance, as commonly applied in prior depression prediction studies (Zulfiker et al., 2021; Vu et al., 2025).

An 80/20 stratified train-test split was applied to preserve class distribution and to support unbiased model evaluation.

In addition, 5-fold CV was applied to reduce variance in performance estimates and assess generalisability across different data partitions. Performance consistency between CV and test results was used to evaluate potential overfitting.

Calibration analysis was conducted to evaluate the reliability of predicted probabilities addressing limitations in prior work that rely primarily on accuracy-based metrics in healthcare settings (Etemadi and Khashei, 2022). Learning curves were used to assess model behaviour under increasing training sizes and to identify overfitting or underfitting, as recommended in recent work

(Deng et al., 2025). Confusion matrices were also used to analyse classification errors, with particular emphasis on false negatives due to their importance in mental health risk detection.

4. Results

4.1 Model Evaluation

The performance of the classification models is summarised in Table 1. ROC-AUC values (~0.92) indicate strong performance, likely influenced by dataset structure and predictive variables such as suicidal ideation. As a result, the performance may reflect dataset signal strength rather than generalisable real-world predictive power.

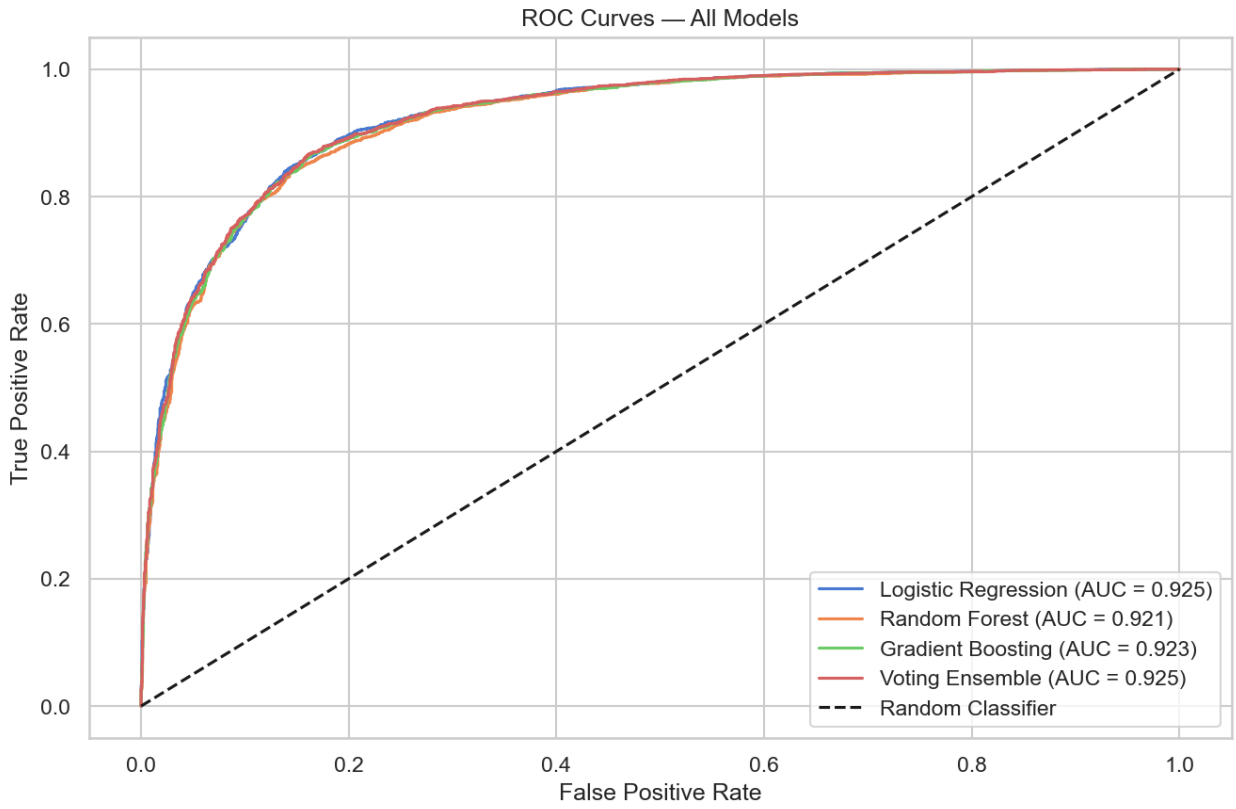
LR consistently ranked at the top across evaluation metrics, achieving an accuracy of 0.8563 and ROC-AUC of 0.9247.

RF, GB, and the VE produced nearly identical results, with performance differences below 1% across all evaluation metrics. This narrow range suggests that increased model complexity does not significantly improve performance.

Overall, the results indicate that depression can be effectively classified using structured student data, with simpler linear models performing competitively against more complex ensemble approaches.

Model	Accuracy	Precision (Depressed)	Recall (Depressed)	F1-Score (Depressed)	ROC-AUC
Logistic Regression	0.8563	0.8658	0.8931	0.8792	0.9247
Random Forest	0.8479	0.8574	0.8879	0.8724	0.9209
Gradient Boosting	0.8531	0.8631	0.8903	0.8765	0.9231
Voting Ensemble	0.8543	0.8634	0.8925	0.8777	0.9246

Table 1: Comparison of classification performance (accuracy, precision, recall, F1-score, and ROC-AUC) for all models on the test set.



4.2 Feature Analysis and Interpretability

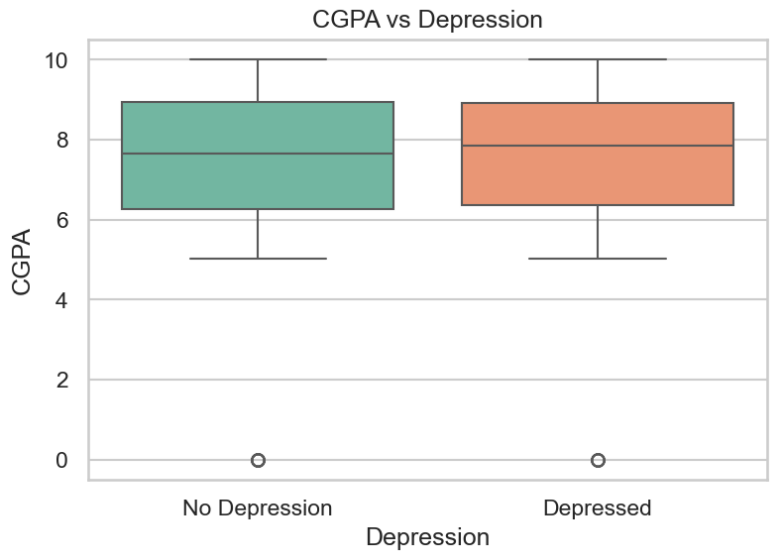
Correlation and SHAP analysis were used to identify the most influential predictors of depression. Correlation results showed that suicidal ideation, academic pressure, and financial stress had the strongest associations with depression, while variables such as CGPA, gender, and family history showed minimal or no meaningful relationship (Appendix C).

SHAP analysis confirmed these findings, demonstrating that a small subset of features dominates model predictions (Appendix D). Suicidal ideation emerged as the most influential feature, followed by academic pressure and financial stress, jointly accounting for over 60% of total model influence. However, these results reflect model contribution strength rather than causal influence. Other variables such as CGPA and gender showed consistently low contribution across all models.

Feature importance analysis from RF further supports these findings, showing that the same core stress-related variables dominate predictive power (0.36, 0.24, 0.13 respectively) (Appendix E).

Boxplot analysis also highlights clear separation between depressed and non-depressed students for key psychological variables, particularly academic pressure and financial stress.

Boxplot analysis (Appendix F) indicates that CGPA shows minimal separation between depressed and non-depressed groups, consistent with its low predictive importance across all models. This suggests that the lack of statistical separation directly corresponds to its minimal role in model-based feature importance, reinforcing its limited relevance in predicting depression within this dataset.



Depression prediction in this dataset is primarily associated with psychological stress variables, which consistently contribute the highest predictive importance across correlation, SHAP, and feature importance analyses. However, these results reflect statistical contribution to model predictions rather than causal relationships.

4.3 Classification Behaviour

Model stability was assessed using CV (Appendix G) and confusion matrix analysis. All models demonstrated stable performance across folds, with minimal variation in ROC-AUC scores, indicating strong generalisation and low sensitivity to variation.

Confusion matrix analysis showed a slight recall bias, meaning models prioritised identifying depressed students over minimising false positives (Appendix H). LR showed the strongest recall performance, resulting in the lowest false negative rate, correctly identifying 2,915 depressed cases while misclassifying 349 instances.

This is appropriate in mental health contexts where false negatives are more critical than false positives.

4.4 Further Evaluation

Interaction analysis indicated a strong combined effect between academic pressure and financial stress (Appendix I). Depression rates increased from ~7% to 95% at high stress levels, indicating strong interaction effects. This shows combined stress factors amplify risk beyond additive effects.

Learning curve analysis showed that LR converged quickly with low variance (Appendix J). RF and GB initially exhibited signs of mild overfitting, but this gap reduced as the dataset size increased. All models plateaued at approximately 20,000 samples, suggesting diminishing returns from additional data.

Calibration analysis indicated that all models produced reliable probability estimates (Appendix K). LR and GB were closest to ideal calibration, while RF showed slight under-confidence in the mid-probability range (0.2–0.5).

A regression analysis examining the relationship between key predictors and CGPA showed minimal explanatory power, suggesting that depression and academic performance are weakly related (Appendix L).

4.5 Discussion

Depression prediction in this dataset is largely driven by simple, structured relationships. The strong performance of LR suggests that most predictive signal is captured through linear combinations of key variables, supporting findings from Vu et al. (2025), who similarly observed competitive performance of linear models in depression prediction tasks.

The minimal performance gap between models highlights diminishing returns from increased complexity, reinforcing that performance differences are driven more by dataset characteristics than model choice (Narayanrao and Lalitha Surya Kumari, 2020). While ensemble methods capture additional interactions, these do not translate into meaningful gains in predictive performance, a limitation also observed in studies using structured and region-specific datasets (Zulfiker et al., 2021), suggesting that flexible models are unnecessary.

The consistent importance of suicidal ideation, academic pressure, and financial stress across all methods reinforces the central role of psychological stressors. The interaction heatmap shows that academic and financial stress compound risk, with depression rising from 7% to 95%, supporting dual-stressor interventions.

Overall, the findings support the use of simpler, interpretable models as an effective balance between accuracy, stability, and transparency, extending prior research by demonstrating that increased model complexity does not necessarily yield practical benefits in structured mental health prediction tasks (Vu et al., 2025).

4.6 Limitations

Despite strong performance, several limitations should be noted. First, the dataset is limited to Indian universities, restricting generalisability across different populations. Second, reliance on self-reported data introduces potential bias in key variables such as stress and suicidal ideation.

Third, the binary classification of depression simplifies a complex clinical condition and does not capture severity. Finally, high model performance may reflect dataset-specific structure rather than real-world robustness, particularly given the strong predictive signal in a small number of features.

5. Conclusion

This study applied multiple machine learning models to predict student depression using a large dataset of 27,901 records. All models achieved strong performance, with ROC-AUC values above 0.92.

LR performed as well as, or slightly better than, more complex models, indicating that predictive relationships are primarily linear. As a result, increased model complexity does not provide meaningful performance gains.

Suicidal ideation, academic pressure, and financial stress were consistently identified as the most important predictors, highlighting the central role of psychological stressors.

Interaction effects did not substantially improve performance. These findings suggest that interpretable models provide an effective balance between accuracy and transparency for early-warning systems.

However, results may be dataset-specific, and further validation across populations is required to assess generalisability.

6. Bibliography

Aceijas, C. *et al.* (2017) 'Determinants of health-related lifestyles among university students', *Perspectives in Public Health*, 137(4), pp. 227–236. Available at: https://doi.org/10.1177/1757913916666875.

Deng, J. *et al.* (2025) 'So You've Got a High AUC, Now What? An Overview of Important Considerations when Bringing Machine-Learning Models from Computer to Bedside', *Medical Decision Making*, 45(6), pp. 640–653. Available at: https://doi.org/10.1177/0272989X251343082.

Etemadi, S. and Khashei, M. (2022) 'Accuracy versus reliability-based modelling approaches for medical decision making', *Computers in Biology and Medicine*, 141, p. 105138. Available at: https://doi.org/10.1016/j.compbiomed.2021.105138.

Klinkman, M.S. *et al.* (1998) 'False Positives, False Negatives, and the Validity of the Diagnosis of Major Depression in Primary Care', *Archives of Family Medicine*, 7(5), p. 451. Available at: https://doi.org/10.1001/archfami.7.5.451.

Lavazza, L. and Morasca, S. (2023) 'Common Problems With the Usage of F-Measure and Accuracy Metrics in Medical Research', *IEEE Access*, 11, pp. 51515–51526. Available at: https://doi.org/10.1109/ACCESS.2023.3278996.

Lundberg, S.M. and Lee, S.-I. (2017) 'A Unified Approach to Interpreting Model Predictions', *Advances in Neural Information Processing Systems*. Curran Associates, Inc. Available at: https://proceedings.neurips.cc/paper_files/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html (Accessed: 11 April 2026).

Lynam, A.L. *et al.* (2020) 'Logistic regression has similar performance to optimised machine learning algorithms in a clinical setting: application to the discrimination between type 1 and type 2 diabetes in young adults', *Diagnostic and Prognostic Research*, 4(1), p. 6. Available at: https://doi.org/10.1186/s41512-020-00075-2.

Narayanrao, P.V. and Lalitha Surya Kumari, P. (2020) 'Analysis of Machine Learning Algorithms for Predicting Depression', *2020 International Conference on Computer Science, Engineering and Applications (ICCSEA)*, pp. 1–4. Available at: https://doi.org/10.1109/ICCSEA49143.2020.9132963.

Silva Filho, T. *et al.* (2023) 'Classifier calibration: a survey on how to assess and improve predicted class probabilities', *Machine Learning*, 112(9), pp. 3211–3260. Available at: https://doi.org/10.1007/s10994-023-06336-7.

Vu, T. *et al.* (2025) 'Prediction of depressive disorder using machine learning approaches: findings from the NHANES', *BMC Medical Informatics and Decision Making*, 25(1), p. 83. Available at: https://doi.org/10.1186/s12911-025-02903-1.

Zulfiker, Md.S. *et al.* (2021) 'An in-depth analysis of machine learning approaches to predict depression', *Current Research in Behavioral Sciences*. Available at: https://doi.org/10.1016/j.crbeha.2021.100044.

7. Appendices

A. Dataset

Variable	Type	Values / Range	Category
Age	Numerical	Discrete integer (years)	Demographic
CGPA	Numerical	Continuous (0–10 scale)	Academic
Work/Study Hours	Numerical	Discrete integer (hours/day)	Behavioural
Academic Pressure	Ordinal	Scale 1–5 (low → high)	Psychological
Work Pressure	Ordinal	Scale 0–5 (low → high)	Occupational
Study Satisfaction	Ordinal	Scale 1–5 (low → high)	Psychological
Job Satisfaction	Ordinal	Scale 0–5 (low → high)	Occupational
Financial Stress	Ordinal	Scale 1–5 (low → high)	Socioeconomic
Sleep Duration	Ordinal	<5, < 5–6, < 7–8 hours	Behavioural
Dietary Habits	Ordinal	Healthy < Moderate < Unhealthy	Behavioural
Gender	Categorical	Male, Female	Demographic
City	Categorical	Multiple cities (nominal categories)	Demographic
Profession	Categorical	Student	Occupational
Degree	Categorical	e.g. BSc, BCA, MSc, PhD	Academic
Suicidal Thoughts	Binary	0 = No, 1 = Yes	Psychological
Family History of Mental Illness	Binary	0 = No, 1 = Yes	Medical History
Depression (Target Variable)	Binary	0 = Not Depressed, 1 = Depressed	Target Variable
Original Dataset		student-depression-dataset.csv.csv	
Cleaned Dataset		student_depression_cleaned_readable.csv	
Encoded Dataset		student_depression_cleaned_encoded.csv	

B. Python Input Code

a. Import

All required libraries are imported at the top of the notebook, covering data manipulation, visualisation, machine learning models, evaluation metrics, and explainability tools.

```
import os
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats

from sklearn.model_selection import train_test_split, cross_val_score, StratifiedKFold, GridSearchCV, learning_curve
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.linear_model import LogisticRegression, LinearRegression
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier, VotingClassifier
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix
from sklearn.metrics import roc_auc_score
from sklearn.metrics import roc_curve
from sklearn.metrics import ConfusionMatrixDisplay
from sklearn.metrics import precision_recall_curve
from sklearn.metrics import average_precision_score
from sklearn.decomposition import PCA
from sklearn.pipeline import Pipeline
from sklearn.calibration import calibration_curve
from scipy.stats import pointbiserialr

import warnings
warnings.filterwarnings('ignore')

os.makedirs("figures", exist_ok=True)

sns.set_theme(style="whitegrid", palette="muted")
plt.rcParams['figure.figsize'] = (10, 6)
```

b. Dataset and Data Preparation

The dataset is loaded from CSV and inspected for shape, data types, missing values, and target class distribution. Categorical variables are label-encoded or ordinally mapped, high-cardinality nominal variables are one-hot encoded, and rows with missing values are removed to produce a clean feature matrix of 27,898 records.

```
df = pd.read_csv("student-depression-dataset.csv.csv")

print("Shape:", df.shape)
print("\nColumn Types:\n", df.dtypes)
print("\nMissing Values:\n", df.isnull().sum())
print("\nTarget Distribution:\n", df['Depression'].value_counts(normalize=True).round(3))
df.head()

df_clean = df.copy()

df_clean.drop(columns=['id'], inplace=True)

binary_cols = ['Gender', 'Have you ever had suicidal thoughts ?', 'Family History of Mental Illness']
le = LabelEncoder()
for col in binary_cols:
    df_clean[col] = le.fit_transform(df_clean[col].astype(str))

sleep_order = {'Less than 5 hours': 0, '5-6 hours': 1, '7-8 hours': 2, 'More than 8 hours': 3}
df_clean['Sleep Duration'] = df_clean['Sleep Duration'].map(sleep_order)

diet_order = {'Unhealthy': 0, 'Moderate': 1, 'Healthy': 2}
df_clean['Dietary Habits'] = df_clean['Dietary Habits'].map(diet_order)

df_clean = pd.get_dummies(df_clean, columns=['City', 'Profession', 'Degree'], drop_first=True)

df_clean.dropna(inplace=True)

print("Cleaned shape:", df_clean.shape)
print("Remaining nulls:", df_clean.isnull().sum().sum())
```

c. Feature Definition and Train-Test

The 11 core predictive features are selected, and the dataset is partitioned into an 80/20 stratified train-test split to preserve the class ratio across both sets.

```
core_features = [
    'Age',
    'Academic Pressure',
    'CGPA',
    'Study Satisfaction',
    'Financial Stress',
    'Work/Study Hours',
    'Sleep Duration',
    'Dietary Habits',
    'Gender',
    'Have you ever had suicidal thoughts ?',
    'Family History of Mental Illness'
]

X = df_clean[core_features]
y = df_clean['Depression']

X_raw_train, X_raw_test, y_raw_train, y_raw_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

print("Train size:", X_raw_train.shape)
print("Test size:", X_raw_test.shape)
```

d. Logistic Regression

A logistic regression model is trained within a pipeline that applies standard scaling prior to fitting and evaluated on the held-out test set using classification report and ROC-AUC.

```
lr_pipe = Pipeline([
    ('scaler', StandardScaler()),
    ('lr', LogisticRegression(max_iter=1000, random_state=42))
])
lr_pipe.fit(X_raw_train, y_raw_train)

y_pred_lr = lr_pipe.predict(X_raw_test)
print("=== Logistic Regression ===")
print(classification_report(y_raw_test, y_pred_lr))

lr_probs = lr_pipe.predict_proba(X_raw_test)[:,1]
print(f"ROC-AUC: {roc_auc_score(y_raw_test, lr_probs):.4f}")
```

e. Random Forest

A random forest ensemble of 200 trees with maximum depth 10 is trained and evaluated, with feature importances computed via mean decrease in impurity.

```
rf_plain = RandomForestClassifier(
    n_estimators=200,
    max_depth=10,
    random_state=42,
    n_jobs=-1
)
rf_plain.fit(X_raw_train, y_raw_train)

y_pred_rf = rf_plain.predict(X_raw_test)
print("=== Random Forest ===")
print(classification_report(y_raw_test, y_pred_rf))

rf_probs = rf_plain.predict_proba(X_raw_test)[:,1]
print(f"ROC-AUC: {roc_auc_score(y_raw_test, rf_probs):.4f}")
```

f. Gradient Boosting

A sequential gradient boosting classifier of 200 shallow trees is trained with a learning rate of 0.05 and evaluated on the test set.

```
gb_plain = GradientBoostingClassifier(
    n_estimators=200,
    learning_rate=0.05,
    max_depth=4,
    random_state=42
)
gb_plain.fit(X_raw_train, y_raw_train)

y_pred_gb = gb_plain.predict(X_raw_test)
print("=== Gradient Boosting ===")
print(classification_report(y_raw_test, y_pred_gb))

gb_probs = gb_plain.predict_proba(X_raw_test)[:,1]
print(f"ROC-AUC: {roc_auc_score(y_raw_test, gb_probs):.4f}")
```

g. Voting ensemble

A soft-voting ensemble combining the probability outputs of logistic regression, random forest, and gradient boosting is constructed and evaluated, with the shared model test map defined for use in subsequent evaluation cells.

```
voting_clf = VotingClassifier(
    estimators=[
        ('lr', lr_pipe),
        ('rf', rf_plain),
        ('gb', gb_plain)
    ],
    voting='soft'
)
voting_clf.fit(X_raw_train, y_raw_train)

y_pred_vc = voting_clf.predict(X_raw_test)
print("=== Soft Voting Ensemble ===")
print(classification_report(y_raw_test, y_pred_vc))

vc_probs = voting_clf.predict_proba(X_raw_test)[:,1]
print(f"ROC-AUC: {roc_auc_score(y_raw_test, vc_probs):.4f}")

model_test_map = [
    ('Logistic Regression', lr_pipe, X_raw_test, y_raw_test),
    ('Random Forest', rf_plain, X_raw_test, y_raw_test),
    ('Gradient Boosting', gb_plain, X_raw_test, y_raw_test),
    ('Voting Ensemble', voting_clf, X_raw_test, y_raw_test),
]
```

h. Hyperparameter Tuning

GridSearchCV with 5-fold cross-validation is applied to the random forest across a grid of 12 parameter combinations to identify the optimal tree depth, estimator count, and minimum sample split.

```
param_grid = {
    'n_estimators': [100, 200],
    'max_depth': [5, 10, 15],
    'min_samples_split': [2, 5]
}

grid_search = GridSearchCV(
    RandomForestClassifier(random_state=42, n_jobs=-1),
    param_grid,
    cv=5,
    scoring='roc_auc',
    n_jobs=-1,
    verbose=1
)
grid_search.fit(X_raw_train, y_raw_train)

print("Best Parameters:", grid_search.best_params_)
print(f"Best CV AUC: {grid_search.best_score_:.4f}")

best_rf = grid_search.best_estimator_
best_preds = best_rf.predict(X_raw_test)
best_probs = best_rf.predict_proba(X_raw_test)[:,1]

print("\n=== Tuned Random Forest ===")
print(classification_report(y_raw_test, best_preds))
print(f"ROC-AUC: {roc_auc_score(y_raw_test, best_probs):.4f}")
```

i. Cross-validation

Stratified 5-fold cross-validation is applied to all four models to assess generalisation stability, reporting mean ROC-AUC, standard deviation, and min/max scores across folds.

```
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

models_cv = {
    'Logistic Regression': lr_pipe,
    'Random Forest': rf_plain,
    'Gradient Boosting': gb_plain,
    'Voting Ensemble': voting_clf
}

print("=" * 60)
print("5-FOLD STRATIFIED CROSS-VALIDATION RESULTS")
print("=" * 60)

cv_results = {}
for name, model in models_cv.items():
    scores = cross_val_score(
        model, X, y,
        cv=cv,
        scoring='roc_auc',
        n_jobs=-1
    )
    cv_results[name] = scores
    print(f"{name:<25} "
          f"Mean AUC: {scores.mean():.4f} "
          f"±{scores.std():.4f} "
          f"Min: {scores.min():.4f} "
          f"Max: {scores.max():.4f}")
```

j. Confusion matrices

Confusion matrices are plotted in a 2x2 grid for all four models, showing true positive, true negative, false positive, and false negative counts on the held-out test set.

```
fig, axes = plt.subplots(2, 2, figsize=(12, 10))
axes = axes.flatten()

for i, (name, model, Xt, yt) in enumerate(model_test_map):
    cm = confusion_matrix(yt, model.predict(Xt))
    disp = ConfusionMatrixDisplay(
        cm,
        display_labels=['No Depression', 'Depressed']
    )
    disp.plot(ax=axes[i], colorbar=False, cmap='Blues')
    axes[i].set_title(name)

plt.suptitle("Confusion Matrices — All Models", fontsize=14)
plt.tight_layout()
plt.savefig("figures/confusion_matrices.png", dpi=150, bbox_inches='tight')
plt.show()
```

k. Calibration curves

Calibration curves are plotted against the perfect calibration diagonal to assess whether each model's predicted probabilities are reliable as genuine risk estimates.

```
plt.figure(figsize=(9, 6))
plt.plot([0,1],[0,1],'k--', label='Perfectly Calibrated')

for name, model, Xt, yt in model_test_map:
    probs = model.predict_proba(Xt)[:,1]
    fraction_pos, mean_pred = calibration_curve(yt, probs, n_bins=10)
    plt.plot(mean_pred, fraction_pos, 'o-', label=name)

plt.xlabel("Mean Predicted Probability")
plt.ylabel("Fraction of Positives")
plt.title("Calibration Curves — Are Predicted Probabilities Reliable?")
plt.legend()
plt.tight_layout()
plt.savefig("figures/calibration_curves.png", dpi=150, bbox_inches='tight')
plt.show()
```

l. SHAP Analysis

SHAP is applied to the random forest to compute instance-level feature attributions, producing both a dot summary plot showing direction and magnitude and a bar chart of mean absolute SHAP values.

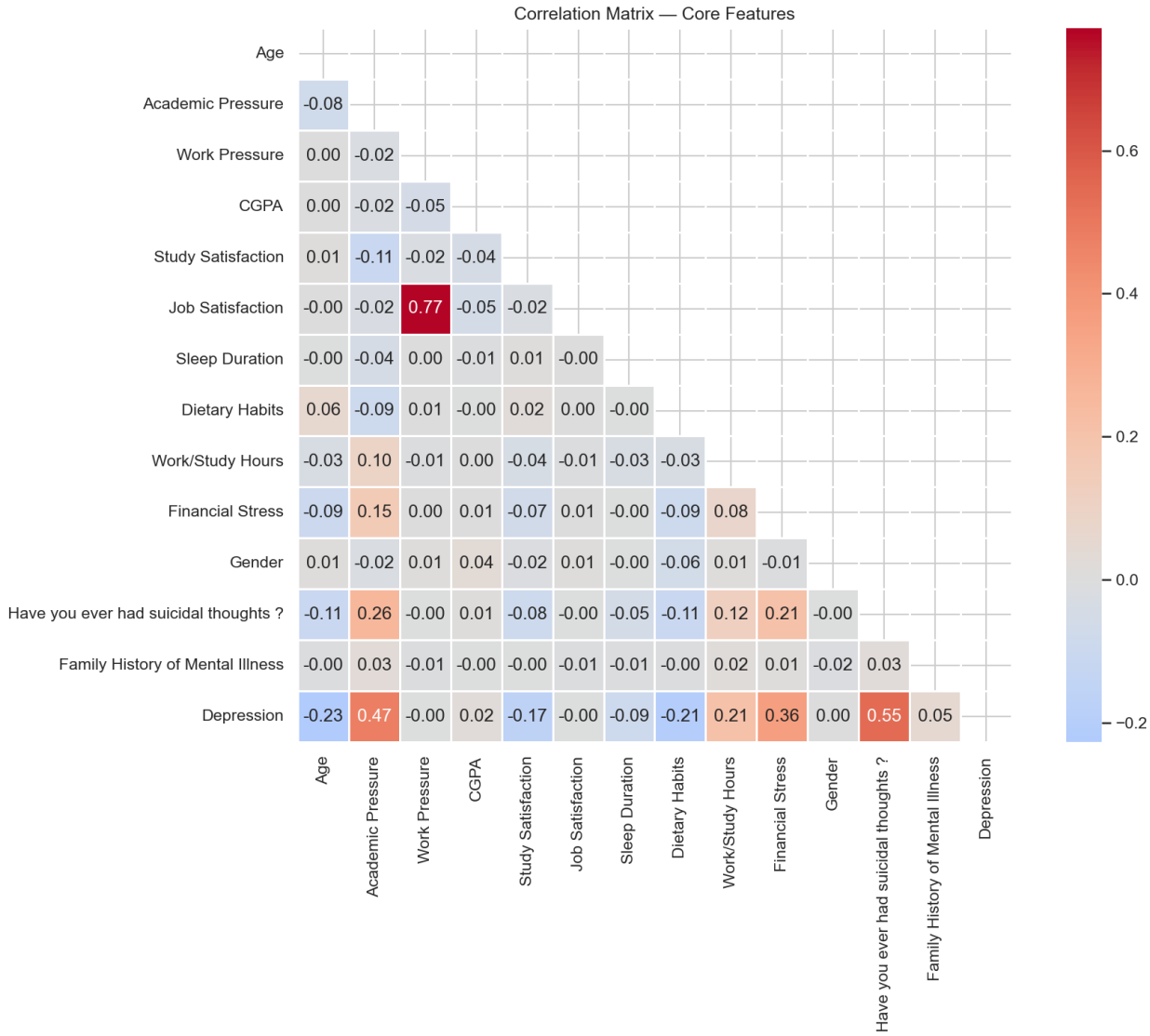
```
import shap

explainer = shap.TreeExplainer(rf_plain)
shap_values = explainer.shap_values(X_raw_test)

shap.summary_plot(
    shap_values[:,:,1],
    X_raw_test,
    feature_names=core_features,
    show=False
)
plt.tight_layout()
plt.savefig("figures/shap_summary.png", dpi=150, bbox_inches='tight')
plt.show()

shap.summary_plot(
    shap_values[:,:,1],
    X_raw_test,
    feature_names=core_features,
    plot_type="bar",
    show=False
)
plt.tight_layout()
plt.savefig("figures/shap_bar.png", dpi=150, bbox_inches='tight')
plt.show()
```

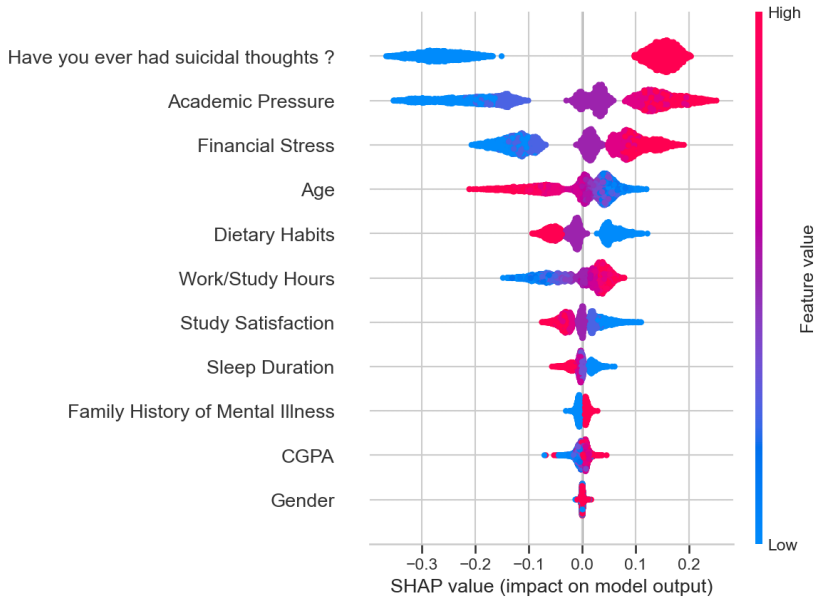
C. Correlation Heatmap



The correlation matrix shows that depression is most strongly associated with suicidal ideation (r = 0.55) and academic pressure (r = 0.47), which appear as the strongest positive relationships. Financial stress (r = 0.36) and work/study hours (r = 0.21) show moderate positive correlations, while age (r = −0.23) and dietary habits (r = −0.21) show moderate negative correlations, indicating higher risk among younger students with poorer lifestyle habits.

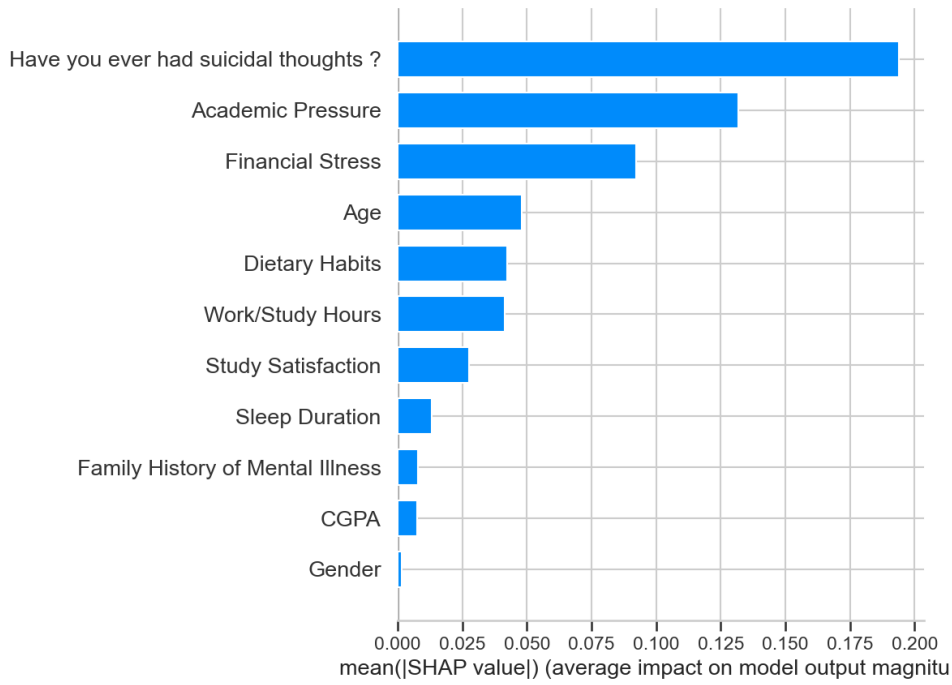
CGPA (r = 0.02) and gender (r = 0.00) show negligible correlation with depression, confirming they are not meaningful predictors. A strong collinearity is also observed between related workload variables, suggesting partial redundancy within the dataset.

D. SHAP Analysis



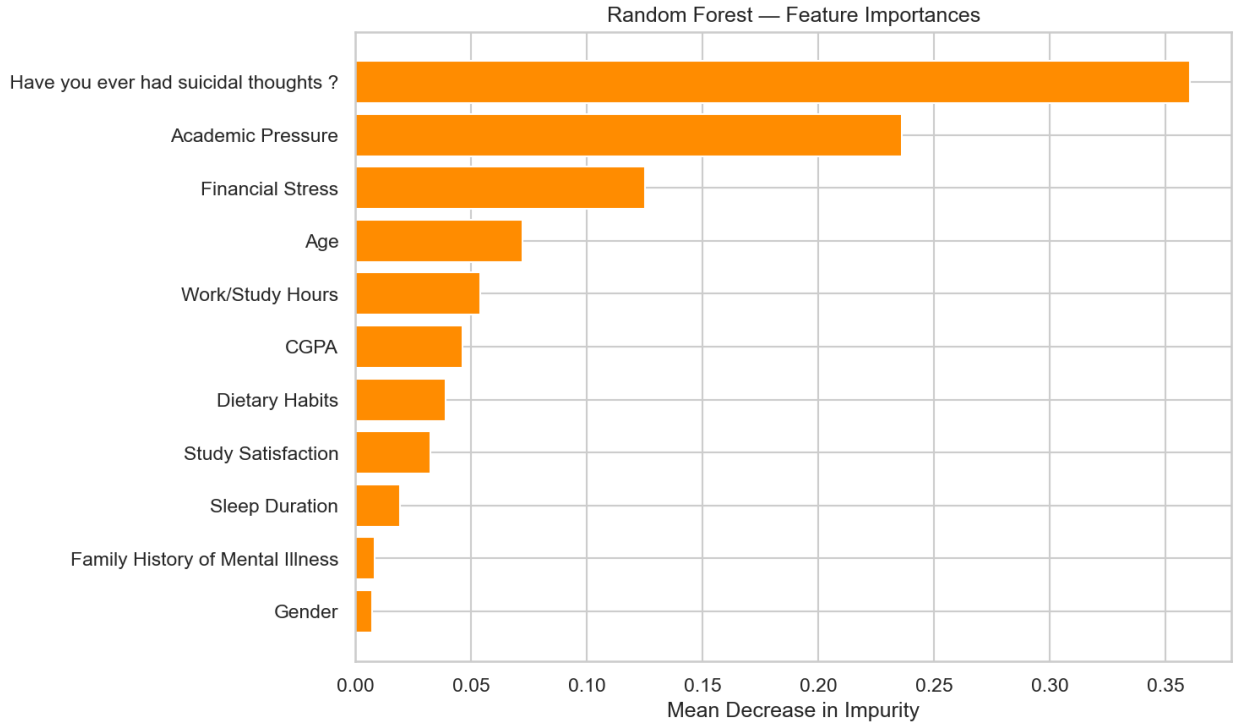
The SHAP summary plot provides insight into both the magnitude and direction of feature effects. Suicidal ideation behaves almost like a binary switch, with strong positive SHAP values when present and strongly negative values when absent. Academic pressure shows a clear monotonic relationship, where higher values consistently increase depression risk, while dietary habits show the opposite effect.

Age demonstrates a protective trend, with older students associated with lower predicted risk. In contrast, gender and family history show negligible SHAP impact, confirming minimal predictive value.



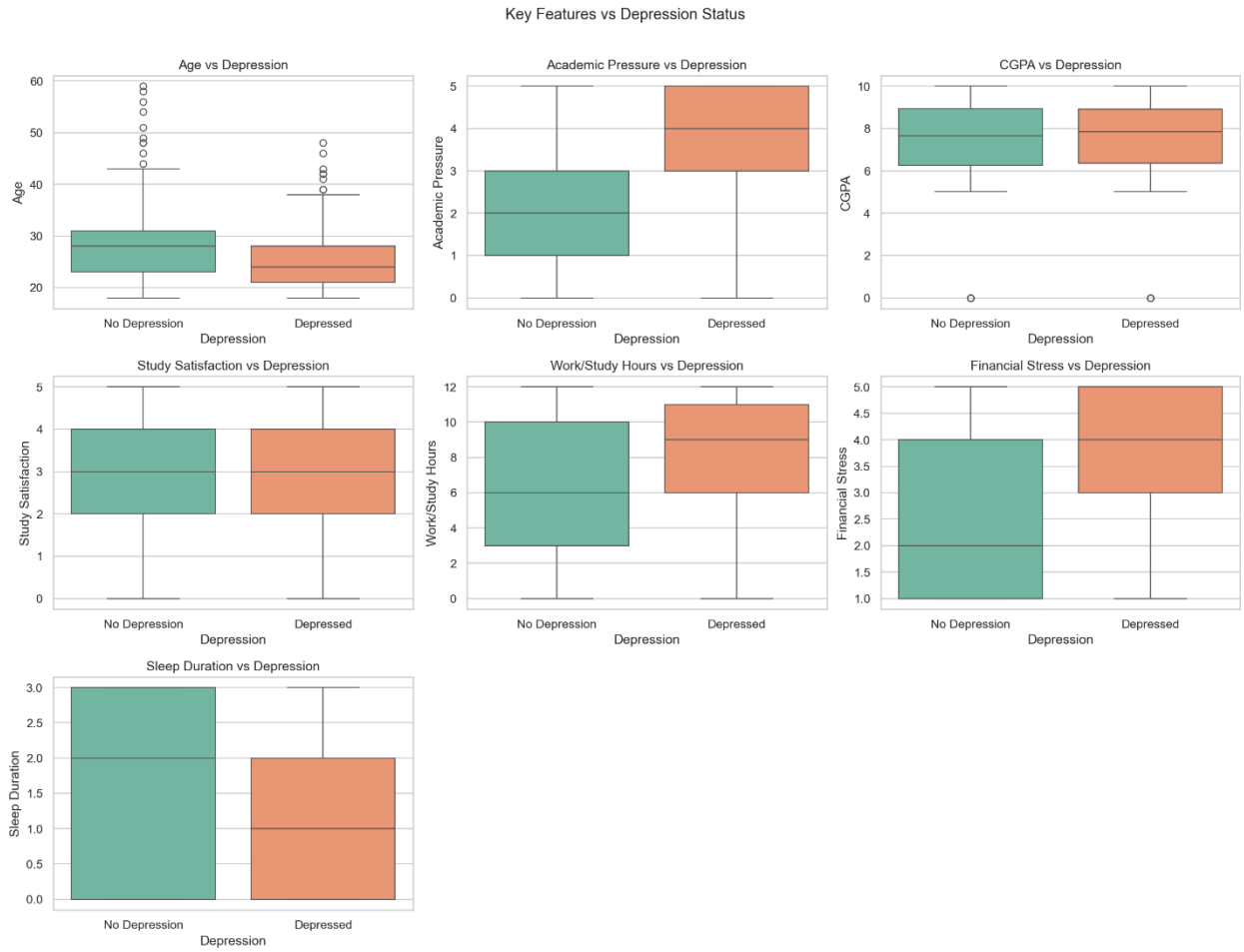
The SHAP bar chart confirms that suicidal ideation (≈0.195), academic pressure (≈0.127), and financial stress (≈0.093) dominate model influence, collectively accounting for over 60% of predictive importance. Other features contribute only marginally, reinforcing that depression is primarily driven by psychological stress factors.

E. Feature Importance using Random Forest



The Random Forest feature importance plot further supports these findings, with suicidal ideation (0.36), academic pressure (0.24), and financial stress (0.13) accounting for approximately 73% of total importance. All other features contribute minimally, with gender near zero.

F. Feature Importance Boxplots



Boxplots show clear separation between depressed and non-depressed students for key variables. Academic pressure and financial stress are significantly higher among depressed students, while age is lower, indicating younger students are more affected. Work/study hours also show a modest upward shift.

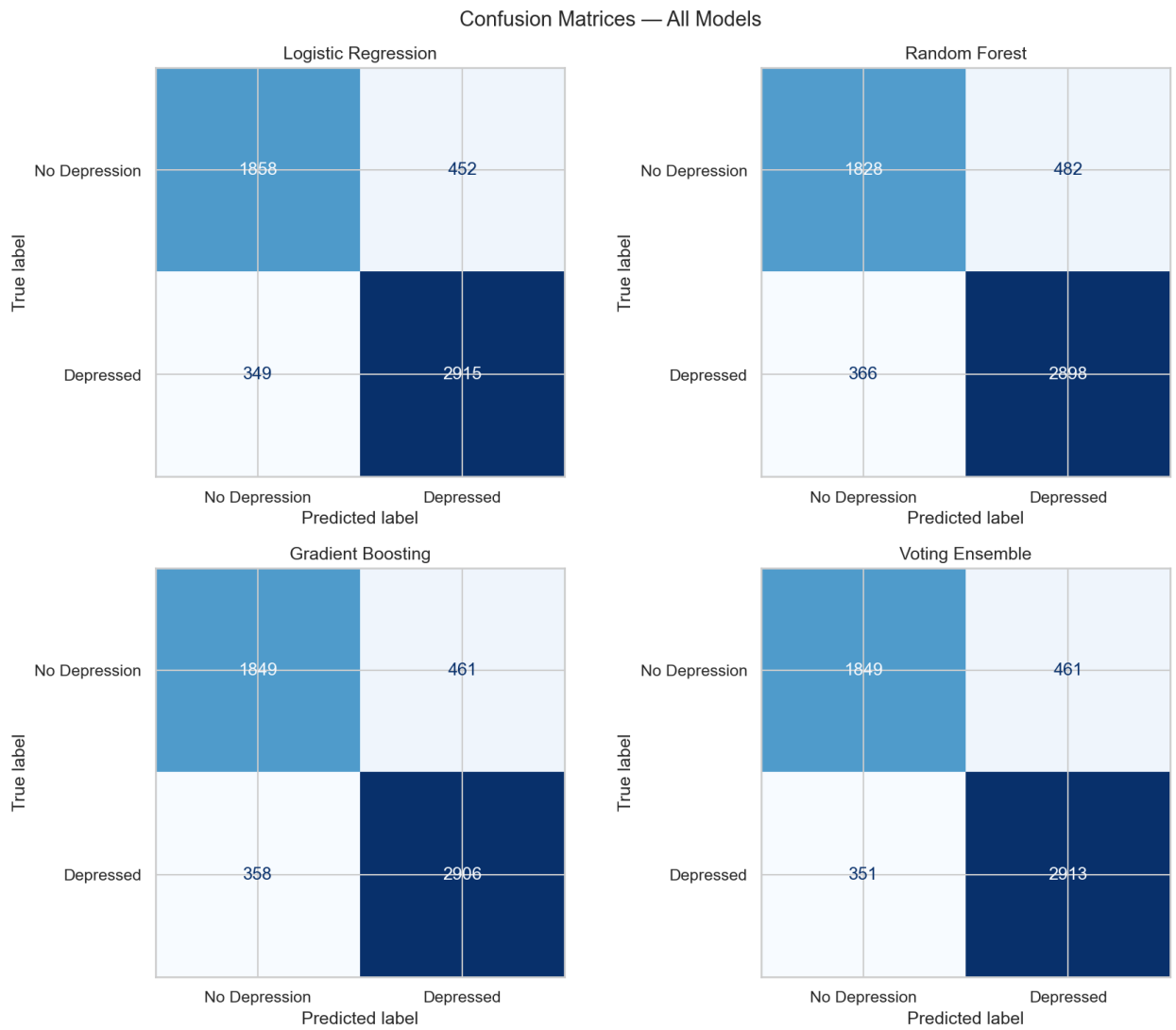
In contrast, CGPA distributions are nearly identical across both groups, with overlapping medians and interquartile ranges, confirming no meaningful relationship between academic performance and depression.

G. Cross-Validation Results

The cross-validation results show that all models generalise consistently across different data splits, with ROC-AUC values tightly clustered between 0.912 and 0.924. The narrow variation indicates stable performance and low sensitivity to training partitions.

Logistic Regression and the Voting Ensemble achieve the highest and most stable medians (~0.922), while Random Forest shows slightly greater variability, including a lower outlier fold (~0.912). Overall, the similarity in distributions suggests that model performance is driven more by the dataset than by algorithmic complexity.

H. Confusion Matrices

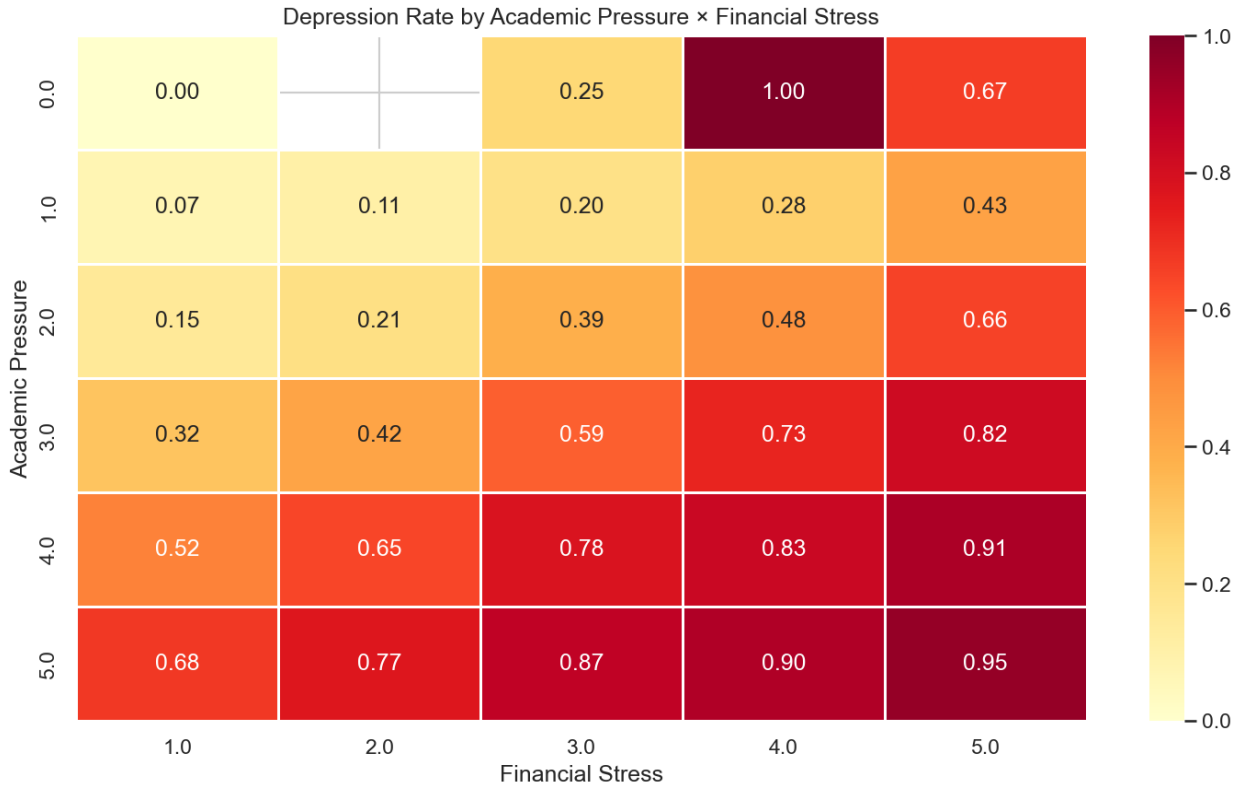


The confusion matrices show consistent classification patterns across all models. Logistic Regression performs best overall, with 349 false negatives and 452 false positives, achieving the lowest rate of missed depression cases.

Random Forest produces slightly higher error counts in both categories, while Gradient Boosting and the Voting Ensemble perform similarly, with marginal differences in false negatives. Across all models, false positives exceed false negatives, indicating a tendency toward recall-oriented classification.

This is appropriate for mental health applications, where failing to identify at-risk individuals is more critical than incorrectly flagging non-depressed students.

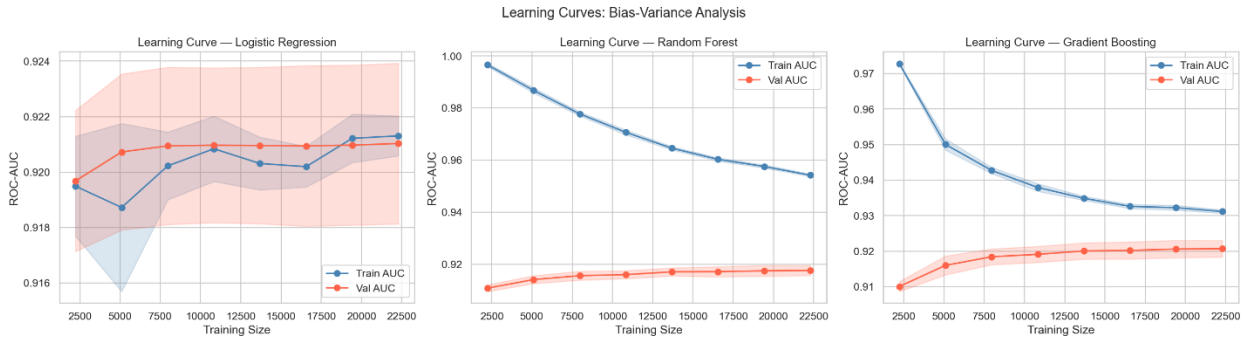
I. Interaction Effects Heatmap



The interaction heatmap reveals a strong compounding effect between academic pressure and financial stress. Depression prevalence increases from approximately 7% at low levels of both variables to 59% at moderate levels, 83% at high levels, and 95% at maximum levels.

This indicates a non-linear interaction where combined stressors significantly amplify risk beyond additive effects. The result suggests that dual-stressor interventions may be more effective than targeting individual factors independently.

J. Learning Curves

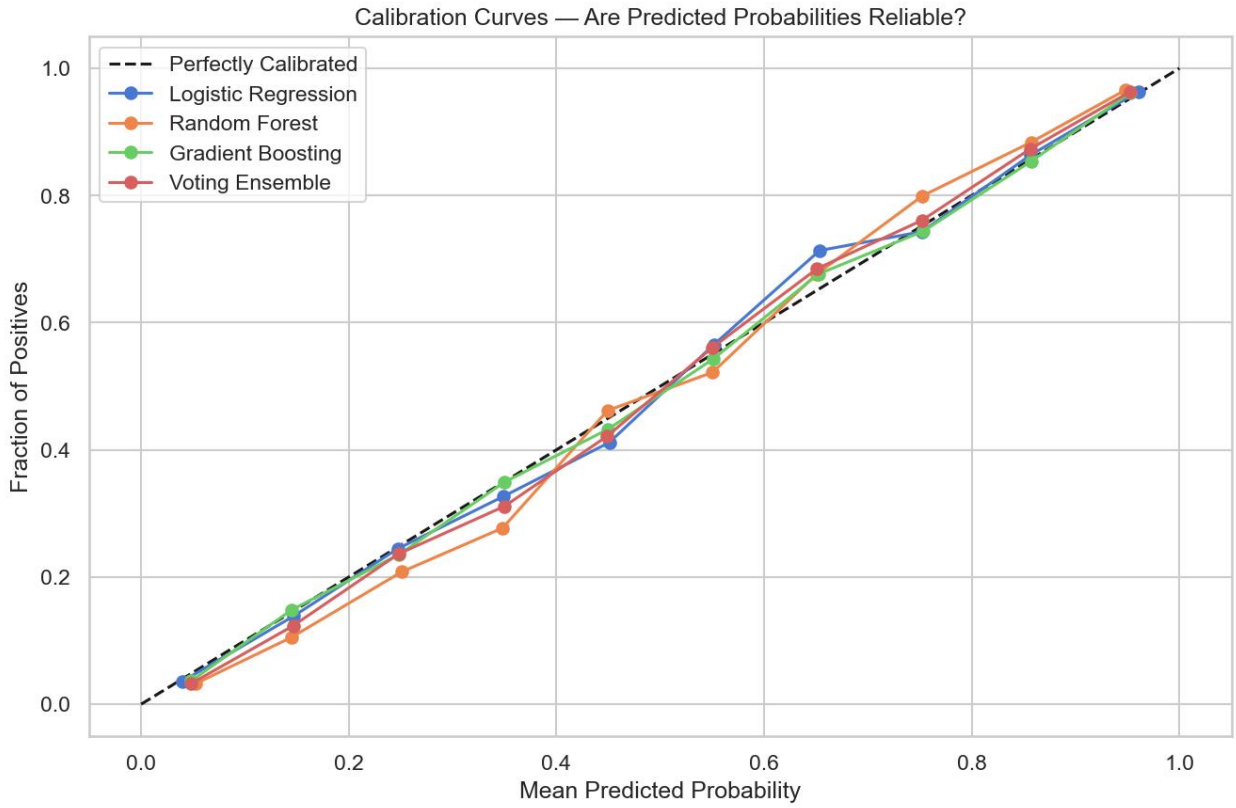


Learning curves highlight differences in model complexity and generalisation behaviour. Logistic Regression shows tightly aligned training and validation performance throughout, stabilising at approximately 0.921, indicating low variance and strong generalisation.

Random Forest initially shows severe overfitting, with near-perfect training performance at small sample sizes, while validation performance remains lower (~0.915). Although this gap reduces with more data, it does not fully close. Gradient Boosting shows a similar but less extreme pattern.

These results indicate that linear models are more stable for this dataset, while ensemble methods benefit more from increased data volume.

K. Calibration Curves

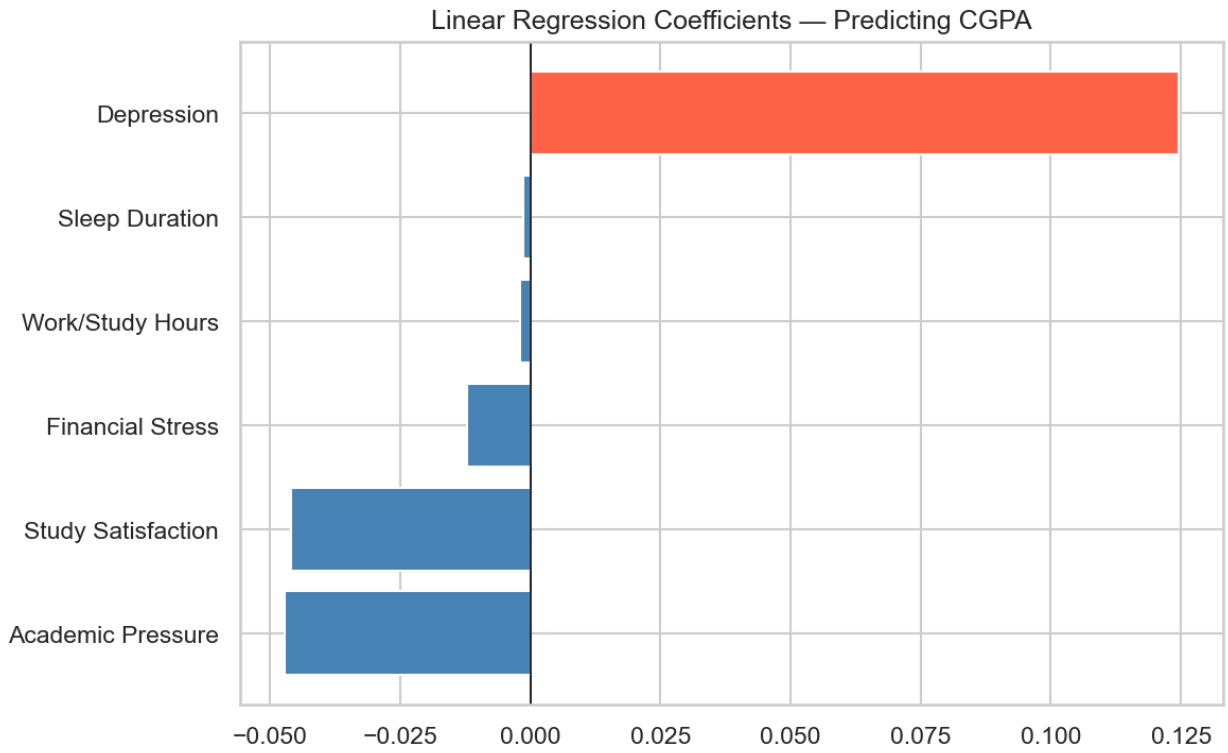


Calibration analysis shows that all models produce well-calibrated probability estimates, closely following the ideal diagonal. This indicates that predicted probabilities correspond well to observed outcomes.

Logistic Regression and Gradient Boosting show the best calibration overall, while Random Forest exhibits slight under-confidence in the mid-probability range (0.2–0.5). The Voting Ensemble performs consistently well across all thresholds.

This confirms that model outputs can be interpreted as reliable risk probabilities, making them suitable for practical screening applications.

L. CGPA Regression



A linear regression model was used to predict CGPA using academic pressure, study satisfaction, work/study hours, financial stress, sleep duration, and depression status. The model achieved an R^2 of 0.003 and RMSE of 1.469, indicating that these variables explain less than 1% of variance in academic performance.

All coefficients were extremely small, with even the largest predictors (depression and academic pressure) showing negligible influence. This confirms that psychological and lifestyle factors associated with depression do not meaningfully predict academic achievement.

These results demonstrate that depression and academic performance are independent outcomes. As a result, academic metrics such as CGPA are insufficient proxies for identifying mental health risk in students.